

STAT 4530/6530

Homework 2

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Course: STAT 6530

Problems

1. Consider the following study done at the National Institute of Science and Technology. Asbestos fibers on filters were counted as part of a project to develop measurement standards for asbestos concentration. Asbestos dissolved in water was spread on a filter, and 3-mm diameter punches were taken from the filter and mounted on a transmission electron microscope. An operator counted the number of fibers in each of 23 grid squares, yielding the following counts:

31 29 19 18 31 28
34 27 34 30 16 18
26 27 27 18 24 22
28 24 21 17 24

We decide to model the counts as arising from a $\text{Poisson}(\mu)$ distribution. The probability mass function for this distribution is:

$$f(y; \mu) = \frac{\mu^y e^{-\mu}}{y!}$$

for $y = 0, 1, 2, \dots$, and the distribution has mean and variance both equal to μ , where the rate parameter μ must be a positive real number.

- (a) (5 points) Find the maximum likelihood estimate of μ . Show your work (don't just write the answer, even though we did this in class).

Solution: Let $Y_1, \dots, Y_n$ be an i.i.d. sample from a $\text{Poisson}(\mu)$ distribution. The likelihood function is given by:

$$L(\mu) = \prod_{i=1}^n \frac{\mu^{y_i} e^{-\mu}}{y_i!} = \frac{\mu^{\sum y_i} e^{-n\mu}}{\prod y_i!}$$

The log-likelihood function is:

$$l(\mu) = \ln L(\mu) = \left(\sum_{i=1}^n y_i \right) \ln(\mu) - n\mu - \sum_{i=1}^n \ln(y_i!)$$

We differentiate with respect to μ and set it to 0 to find the MLE:

$$\frac{d}{d\mu} l(\mu) = \frac{\sum y_i}{\mu} - n = 0$$

$$\frac{\sum y_i}{\mu} = n \implies \hat{\mu} = \frac{\sum_{i=1}^n y_i}{n}$$

This is equal to the population mean. Using the data provided: Sum of counts ($\sum y_i$) = $31 + 29 + \dots + 24 = 573$. Sample size (n) = 23.

$$\hat{\mu} = \frac{573}{23} \approx 24.913$$

- (b) (5 points) Find an approximate 90% confidence interval for μ .

Solution: For a large sample, the sampling distribution of the mean is approximately normal with standard error $\sigma/\sqrt{n}$. For a Poisson distribution, the variance is equal to the mean ($\sigma^2 = \mu$), so we estimate the standard error as $\sqrt{\hat{\mu}/n}$.

The 90% confidence interval corresponds to $z_{0.05} = 1.645$.

$$\text{CI} = \hat{\mu} \pm 1.645 \sqrt{\frac{\hat{\mu}}{n}}$$

$$\text{CI} = 24.913 \pm 1.645 \sqrt{\frac{24.913}{23}}$$

$$\text{CI} = 24.913 \pm 1.645(1.0408)$$

$$\text{CI} \approx [23.20, 26.63]$$

2. Consider an i.i.d. sample of size n from a distribution with probability mass function

$$f(y; p) = p(1 - p)^{y-1}$$

for $y = 1, 2, 3, \dots$, with $0 < p \leq 1$.

- (a) (5 points) Find the maximum likelihood estimate of p .

Solution: The likelihood function is:

$$L(p) = \prod_{i=1}^n p(1 - p)^{y_i-1} = p^n (1 - p)^{\sum y_i - n}$$

The log-likelihood is:

$$l(p) = n \ln(p) + (\sum y_i - n) \ln(1 - p)$$

Differentiating with respect to p and setting the expression to 0:

$$\frac{d}{dp} l(p) = \frac{n}{p} - \frac{\sum y_i - n}{1 - p} = 0$$

$$\frac{n}{p} = \frac{\sum y_i - n}{1 - p}$$

$$n(1 - p) = p(\sum y_i - n)$$

$$n - np = p \sum y_i - np$$

$$p \sum y_i = n \implies \hat{p} = \frac{n}{\sum y_i} = \frac{1}{\bar{y}}$$

- (b) (5 points) Find the Fisher information $\mathcal{I}(p)$.

Solution: The Fisher information for a single observation is given by $\mathcal{I}_1(p) = -E \left[\frac{\partial^2}{\partial p^2} \ln f(Y; p) \right]$.

$$\ln f(Y; p) = \ln(p) + (Y - 1) \ln(1 - p)$$

$$\frac{\partial}{\partial p} \ln f = \frac{1}{p} - \frac{Y - 1}{1 - p}$$

$$\frac{\partial^2}{\partial p^2} \ln f = -\frac{1}{p^2} - \frac{Y - 1}{(1 - p)^2}$$

Taking the negative expectation (We have $E[Y] = 1/p$, because the original function is a geometric distribution):

$$\begin{aligned} \mathcal{I}_1(p) &= \frac{1}{p^2} + \frac{E[Y] - 1}{(1 - p)^2} = \frac{1}{p^2} + \frac{1/p - 1}{(1 - p)^2} \\ &= \frac{1}{p^2} + \frac{1}{p(1 - p)} \\ &= \frac{1 - p + p}{p^2(1 - p)} = \frac{1}{p^2(1 - p)} \end{aligned}$$

For a sample of size n , the Fisher information is $\mathcal{I}_n(p) = n\mathcal{I}_1(p)$:

$$\mathcal{I}_n(p) = \frac{n}{p^2(1 - p)}$$

- (c) (5 points) Find the asymptotic variance (meaning, the approximate variance when the sample size n is large) of the maximum likelihood estimate.

Solution: The asymptotic variance of the MLE is the inverse of the Fisher Information:

$$\text{Var}(\hat{p}) = \frac{1}{\mathcal{I}_n(p)} = \frac{p^2(1 - p)}{n}$$

3. Consider an i.i.d. sample of size n from a $N(\mu, \sigma^2)$ distribution.

- (a) (5 points) Show that the sample mean and sample variance make up a two-dimensional sufficient statistic for (μ, σ^2) .

Solution: The joint pdf of the sample is:

$$\begin{aligned} f(\mathbf{y}; \mu, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(y_i - \mu)^2}{2\sigma^2} \right) \\ &= (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right) \end{aligned}$$

Using the identity $\sum(y_i - \mu)^2 = \sum(y_i - \bar{y})^2 + n(\bar{y} - \mu)^2$:

$$= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum(y_i - \bar{y})^2}{2\sigma^2} - \frac{n(\bar{y} - \mu)^2}{2\sigma^2}\right)$$

By the Factorization Theorem, we can write $f(\mathbf{y}) = g(T(\mathbf{y}); \theta) \cdot h(\mathbf{y})$, where $h(\mathbf{y}) = 1$. The function g depends on the data only through $\bar{y}$ (sample mean) and $\sum(y_i - \bar{y})^2$ (which is $(n-1)s^2$). Thus, $(\bar{Y}, S^2)$ is sufficient for (μ, σ^2) . Even without the Factorization theorem, we can see that the first term in the exponent is a multiple of the variance, and hence, this entire expression can be estimated with just the sample mean and variations alone.

- (b) (5 points) Suppose we know that $\sigma^2 = 4$ but μ unknown. Find a sufficient statistic for μ .

Solution: Substituting $\sigma^2 = 4$ into the joint pdf:

$$\begin{aligned} f(\mathbf{y}; \mu) &= 4\sqrt{2\pi}^{-n/2} \exp\left(-\frac{1}{8} \sum(y_i - \mu)^2\right) \\ &= 4\sqrt{2\pi}^{-n/2} \exp\left(-\frac{1}{8}(\sum y_i^2 - 2\mu \sum y_i + n\mu^2)\right) \\ &= 4\sqrt{2\pi}^{-n/2} \exp\left(-\frac{\sum y_i^2}{8}\right) \cdot \exp\left(\frac{\mu \sum y_i}{4} - \frac{n\mu^2}{8}\right) \end{aligned}$$

Using the Factorization Theorem, let $h(\mathbf{y}) = \exp(-\sum y_i^2/8)$ and $g(T(\mathbf{y}); \mu) = \exp(\frac{\mu \sum y_i}{4} - \frac{n\mu^2}{8})$. The part depending on μ depends on the data only through $\sum Y_i$. Therefore, $T = \sum_{i=1}^n Y_i$ (or equivalently $\bar{Y}$) is a sufficient statistic for μ . Even without the factorization theorem, we can see, that to ascertain μ , we only need the sample mean and expressions utilizing it.

- (c) (5 points) Suppose we know that $\mu = -3$ but σ^2 is unknown. Find a sufficient statistic for σ^2 .

Solution: We substitute $\mu = -3$ into the joint pdf:

$$\begin{aligned} f(\mathbf{y}; \sigma^2) &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (-3))^2\right) \\ &= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i + 3)^2\right) \end{aligned}$$

Here, using the Factorization Theorem, the function depends on the data only through the term $\sum_{i=1}^n (y_i + 3)^2$. Thus, $T = \sum_{i=1}^n (Y_i + 3)^2$ is a sufficient statistic for σ^2 .

4. (10 points) A social scientist wanted to estimate the proportion of school children in Boston who live in a single-parent family. She decided to use a sample size such that, with probability 0.95, the error would not exceed 0.05. How large a sample size should she use, if she has no idea of the size of that proportion?

Solution: We want a 95% confidence interval with a margin of error $E = 0.05$. The formula for sample size is:

$$n = \frac{z_{\alpha/2}^2 \cdot p(1-p)}{E^2}$$

For a 95% confidence level, $z_{\alpha/2} = 1.96$. Since we have no idea of the size of the proportion p , we use the value of p which maximizes the variance $p(1-p)$ to ensure the sample size is sufficient for any p . This maximum occurs at $p = 0.5$.

$$n = \frac{(1.96)^2 \cdot 0.5(1-0.5)}{(0.05)^2}$$

$$n = \frac{3.8416 \cdot 0.25}{0.0025} = 384.16$$

We round this up, arrive at a sample size of $n = 385$.

5. Consider collecting a sample of size $n = 1497$ from a population with the goal of estimating the proportion of the population that prefers coffee rather than tea. Suppose that the true proportion is $p = 0.53$.

- (a) (10 points) Use R to to simulate s different samples of size n from the population (using the true value of p to simulate the data). For each sample, construct a 95% score confidence interval for p . Report the percentage of the s confidence intervals that contain the true value of p . Do this for $s = 5, 10, 100, 1000$.

Solution: The R code below performs the simulation using `prop.test` to generate Score Confidence intervals without continuity correction.

```
set.seed(123)
run_simulation <- function(s, confidence_level) {
  n <- 1497
  p_true <- 0.53
  coverage_count <- 0

  for (i in 1:s) {
    successes <- rbinom(1, n, p_true)

    # We calculate Score Interval using prop.test
    test_result <- prop.test(successes, n, conf.level = conf_
      level, correct = FALSE)
    ci <- test_result$conf.int

    if (p_true >= ci[1] && p_true <= ci[2]) {
      coverage_count <- coverage_count + 1
    }
  }
  return((coverage_count / s) * 100)
}
# For s = 5, 10, 100, 1000 with 95% confidence
```

```
s_values <- c(5, 10, 100, 1000)
cat("Coverage for 95% Score Intervals:\n")
for (s in s_values) {
  coverage <- run_simulation(s, 0.95)
  cat(sprintf("s = %d: %.1f%%\n", s, coverage))
}
```

Coverage for 95 % Score Intervals:

- s = 5: 80.0%
- s = 10: 80.0%
- s = 100: 96.0%
- s = 1000: 95.4%

- (b) (10 points) Repeat part [(a)], but this time construct 70% score confidence intervals.

Solution:

```
#With 70% confidence
cat("\nCoverage for 70% Score Intervals:\n")
for (s in s_values) {
  coverage <- run_simulation(s, 0.70)
  cat(sprintf("s = %d: %.1f%%\n", s, coverage))
}
```

Coverage for 70 % Score Intervals:

- s = 5: 80.0%
- s = 10: 90.0%
- s = 100: 70.0%
- s = 1000: 71.2%

6. Consider a sample $\mathbf{Y} = (Y_1, \dots, Y_n)$ that we wish to use to estimate the parameter θ . Suppose that $\theta_1(\mathbf{Y})$ is an estimator of θ with $E[(\theta_1(\mathbf{Y}))^2] < \infty$ for all θ , suppose that $T(\mathbf{Y})$ is a sufficient statistic for θ , and let $\theta_2(\mathbf{Y}) = E[\theta_1(\mathbf{Y})|T(\mathbf{Y})]$. Then, for all θ ,

$$E[(\theta_2(\mathbf{Y}) - \theta)^2] \leq E[(\theta_1(\mathbf{Y}) - \theta)^2],$$

and the inequality is strict unless $\theta_1(\mathbf{Y}) = \theta_2(\mathbf{Y})$.

- (a) (5 points) Note that $\theta_1(\mathbf{Y})$ and $\theta_2(\mathbf{Y})$ are both estimators of θ . Interpret the statement above in the context of comparing the two estimators and what it implies about using sufficient statistics to construct estimators.

Solution: It implies that by conditioning an estimator θ_1 on a sufficient statistic T , we obtain a new estimator θ_2 that has the same expectation, but has a variance that is less than or equal to the original. Hence, we can conclude that optimal estimators should be functions of sufficient statistics.

- (b) **This question is only required for students enrolled in STAT 6530.** (10 points) Provide a proof of the statement above. Hints: First, show that the two estimators have the same means, so it suffices to compare their variances. To compare

the variances of the estimators, recall the following result: if X and Z are random variables and X has finite variance, then $Var(X) = E[Var(X|Z)] + Var(E[X|Z])$.

Proof: The Mean Squared Error of an estimator $\hat{\theta}$ is given by:

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + [Bias(\hat{\theta})]^2$$

First, we try and verify the bias. By the Law of Iterated Expectations:

$$E[\theta_2(\mathbf{Y})] = E[E[\theta_1(\mathbf{Y})|T(\mathbf{Y})]] = E[\theta_1(\mathbf{Y})]$$

Since their expected values are equal, their biases are also equal: $Bias(\theta_2) = Bias(\theta_1)$. Therefore, we can say that comparing their MSEs is equivalent to comparing their variances.

Using the conditional variance identity provided in the hint with $X = \theta_1(\mathbf{Y})$ and $Z = T(\mathbf{Y})$:

$$Var(\theta_1) = E[Var(\theta_1|T)] + Var(E[\theta_1|T])$$

Substituting $\theta_2 = E[\theta_1|T]$, we get:

$$Var(\theta_1) = E[Var(\theta_1|T)] + Var(\theta_2)$$

Since we know that variance is always non-negative, $Var(\theta_1|T) \geq 0$, and thus its expectation $E[Var(\theta_1|T)] \geq 0$. hence, we can say:

$$Var(\theta_1) \geq Var(\theta_2)$$

Given the equality of biases, we can conclude:

$$MSE(\theta_2) \leq MSE(\theta_1)$$

This inequality is strict unless $E[Var(\theta_1|T)] = 0$. Since variance is non-negative, this expectation value is zero only if $Var(\theta_1|T) = 0$ with probability 1. If the conditional variance is zero, θ_1 must be a constant given T , meaning θ_1 is a function of T . Since $\theta_2 = E[\theta_1|T]$, and θ_1 is a function of T , then $\theta_1 = \theta_2$. In this case, the inequality no longer is strict.